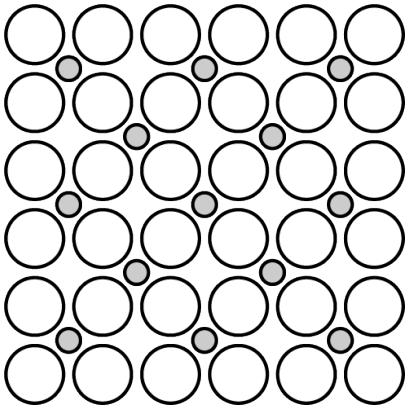
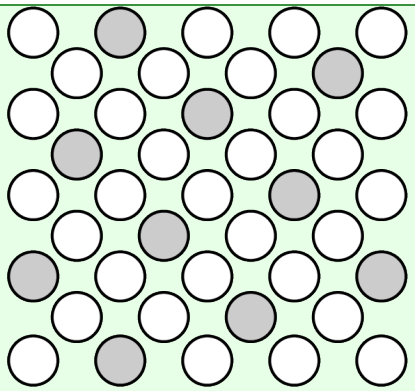
2.4

1. Copper atoms and zinc atoms have the same atomic radius, 135 picometers. Based on this information, which of the following diagrams best represents an alloy containing only copper and zinc atoms?

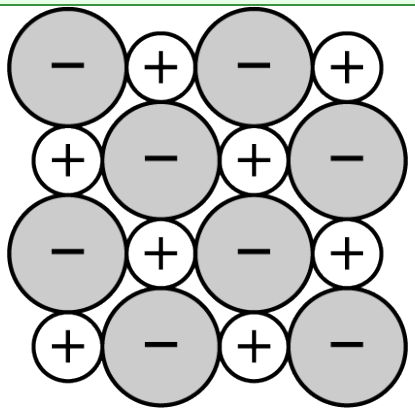
(A)



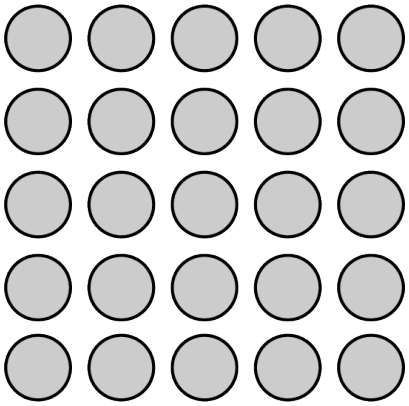
(B) ✓



(C)



(D)

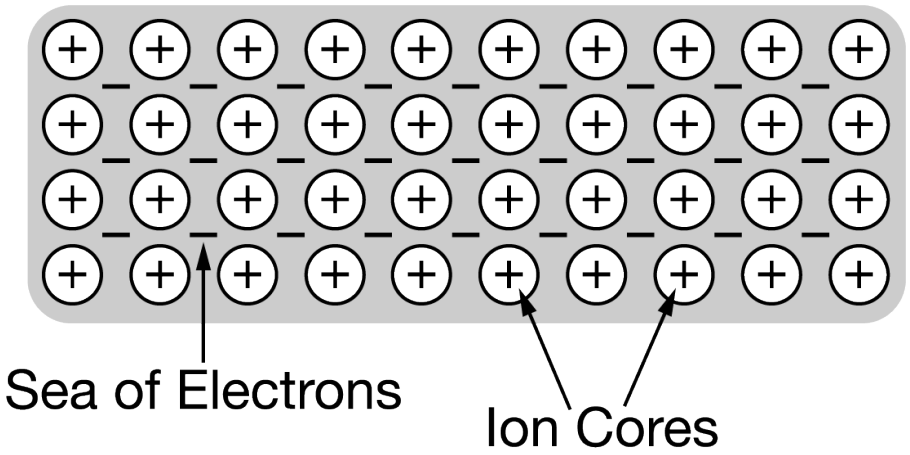


2.4

Answer B

Correct. Being the same size, copper atoms and zinc atoms would form a substitutional alloy in which the atoms substitute for one another in a lattice of positive cores immersed in a sea of electrons.

2.



A particle-level diagram of a metallic element is shown above. Typically, metals are both malleable and ductile. The best explanation for these properties is that the electrons involved in bonding among metal atoms are

(A) unequally shared and form nondirectional bonds

(B) unequally shared and form highly directional bonds

(C) equally shared and form nondirectional bonds ✓

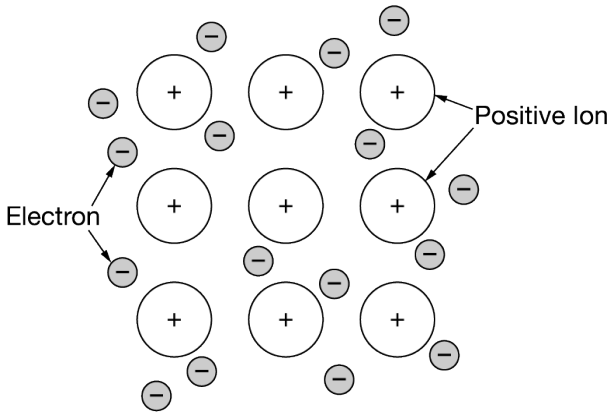
(D) equally shared and form highly directional bonds

Answer C

Correct. In metals, electrons are free to move as described by the electron sea model. Electrons are shared equally among atoms, and the structure is based on simple close-packed structures that contain no fixed directional bonds.

2.4

3.



Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative particles.

(B) It conducts electricity because electrons are free to move through the substance. ✓

(C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.

(D) It does not conduct electricity because the positive particles are not free to move through the substance.

Answer B

Correct. The diagram shows the structure of a metallic solid, which consists of a lattice of positive ions immersed in a sea of electrons in which electrons can move freely.

4.

Element	Metallic Radius (pm)	Melting Point (°C)	Common Oxidation State
Au	144	1064	1+, 3+
Cu	128	1085	1+, 2+
Ag	144	961	1+

To make Au stronger and harder, it is often alloyed with other metals, such as Cu and Ag. Consider two alloys, one of Au and Cu and one of Au and Ag, each with the same mole fraction of Au. If the Au/Cu alloy is harder than the Au/Ag alloy, then which of the following is the best explanation based on the information in the table above?

(A) Cu has two common oxidation states, but Ag has only one.

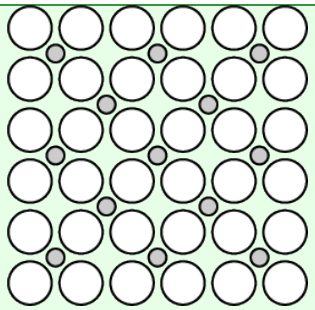
(B) Cu has a higher melting point than Au has, but Ag has a lower melting point than Au has.

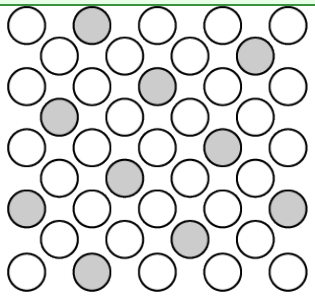
(C) Cu atoms are smaller than Ag atoms, thus they interfere more with the displacement of atoms in the alloy. ✓

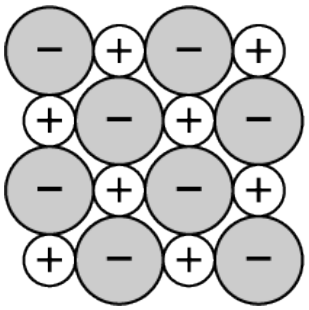
(D) Cu atoms are less polarizable than are Au or Ag atoms, thus Cu has weaker interparticle forces.

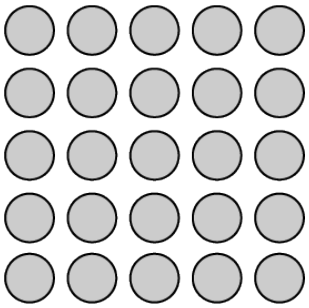
2.4

5. Which of the following particulate-level diagrams best represents an interstitial alloy?

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. In an interstitial alloy, atoms of one of the constituents are significantly smaller than the atoms of the other constituent and occupy the smaller spaces (interstices) among the larger atoms.

2.4

Atomic radii and electronegativity values for some elements are given in the table.

Element	Radius	Electronegativity
Ag	165 pm	1.93
Au	174 pm	2.54
Cu	145 pm	1.90
N	56 pm	3.04
Ni	149 pm	1.91
W	193 pm	2.36

6. Which of the following names a pair of elements that would likely form a substitutional alloy and explains why?

(A) Ni and Cu, because Ni atoms would fit in the spaces between Cu atoms in the solid lattice.

(B) Ni and Cu, because Ni atoms would take the place of Cu atoms in the solid lattice. ✓

(C) W and N, because N atoms would fit in the spaces between W atoms in the solid lattice.

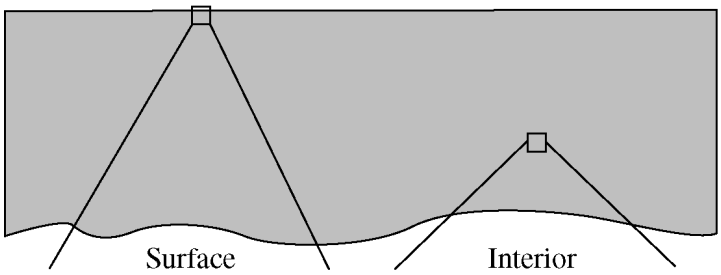
(D) W and N, because N atoms would take the place of W atoms in the solid lattice.

Answer B

Correct. Substitutional alloys form between elements with atoms of comparable radius, where atoms of one element substitute for atoms of another element in the solid lattice. Ni and Cu have similar radii.

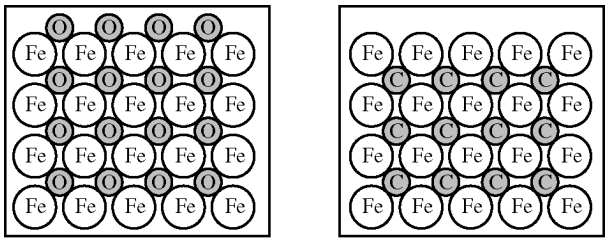
2.4

7. Steel is an alloy consisting of Fe with a small amount of C. Elemental Cr can be added to steel to make the steel less likely to rust; Cr atoms react with oxygen in the air to form a nonreactive layer of chromium oxide on the surface of the steel, preventing the oxidation of underlying Fe atoms. A sample of steel-chromium alloy contains 15 percent Cr by mass. Which of the following diagrams best shows a particle-level view of a surface section and an interior section of the alloy represented below at the left? (The atomic radii of the atoms involved are given in the table below at the right.)

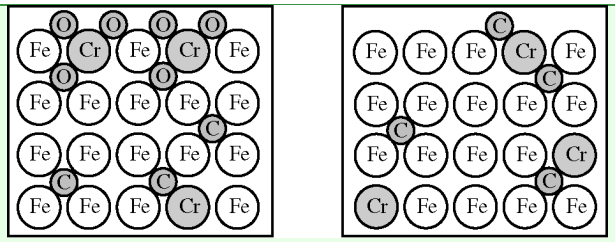


Element	Molar Mass (g/mol)	Atomic Radius (pm)
Fe	55.85	125
Cr	52.00	127
C	12.01	77
O	16.00	73

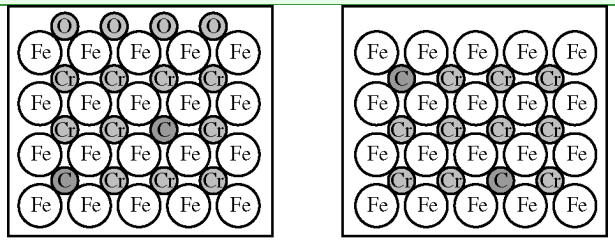
(A)



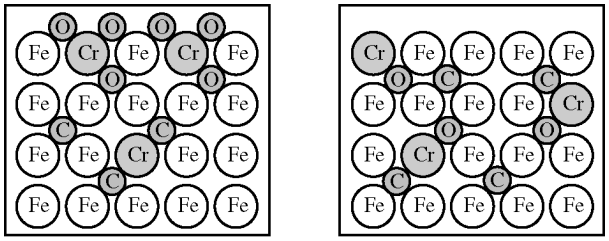
(B) ✓



(C)

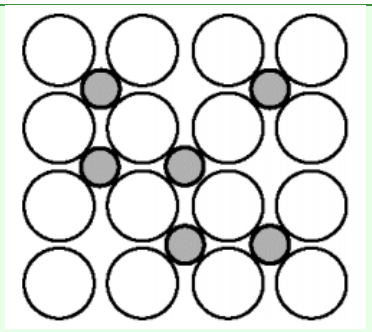


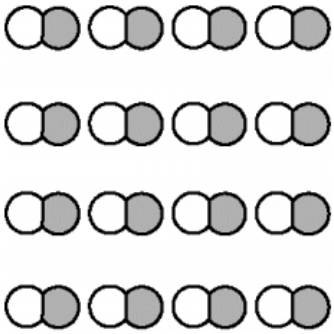
(D)

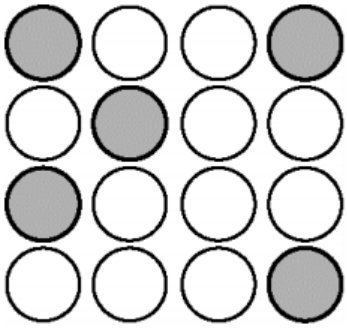


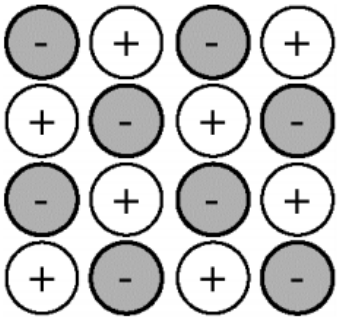
8. Which of the following diagrams best depicts an alloy of Ni and B?

2.4

(A) 

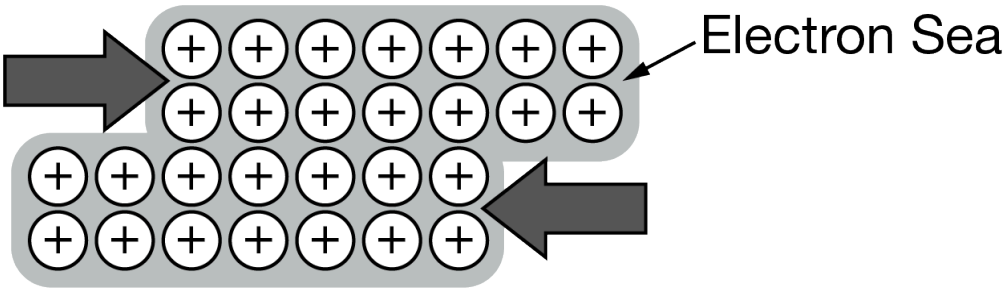
(B) 

(C) 

(D) 

2.4

9.



The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

(A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.

(B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓

(C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.

(D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.

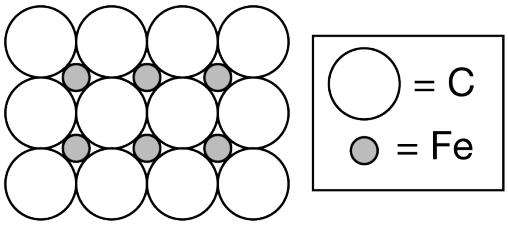
Answer B

Correct. The diagram shows the structure of a metallic solid, which has adjacent layers of positive ions sliding past one another (as indicated by the arrows) while staying within the shading of the electron sea.

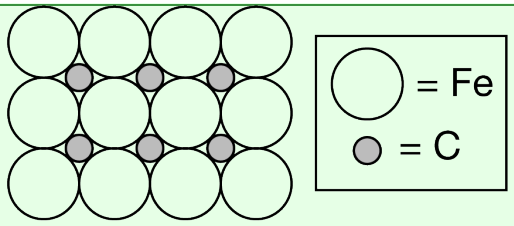
10. Steel is an alloy containing Fe atoms and C atoms. Which of the following diagrams best represents the particle-level structure of steel?

2.4

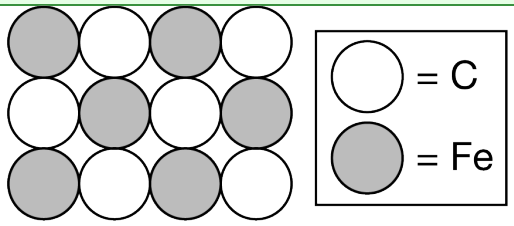
(A)



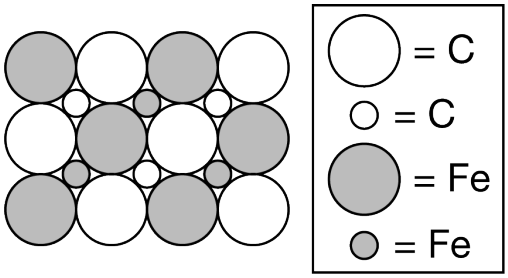
(B) ✓



(C)



(D)



Answer B

Correct. C is in the second period and has two electron shells, whereas Fe is in the fourth period and has four electron shells; thus C atoms are smaller than Fe atoms. The diagram shows how the smaller C atoms occupy the spaces (interstices) between the larger Fe atoms in an interstitial alloy.

11.

Type of Steel	% Carbon	Characteristics	Uses
Low-carbon steel	< 0.2 %	Malleable and ductile	Chains and nails
High-carbon steel	0.6 – 1.5 %	Hard and brittle	Cutting tools

The table above provides some information about two types of steel, both of which are alloys of iron and carbon. Which of the following best helps to explain why high-carbon steel is more rigid than low-carbon steel?

2.4

(A) Elemental carbon is harder than elemental iron.

(B) The additional carbon atoms within the alloy make the high-carbon steel less dense.

(C) The additional carbon atoms within the alloy increase the thermal conductivity of the high-carbon steel.

(D) The additional carbon atoms within the alloy make it more difficult for the iron atoms to slide past one another. ✓